

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XIII

JEE (Main)-2022

TEST DATE: 31-05-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. B
Sol. Distance, acceleration, distance, time

2. B
Sol. According to Ohm's law,

$$V = RA \quad \text{or} \quad R = \frac{V}{A}$$

$$\text{Dimension of } V = \frac{W}{q} = \frac{[ML^2T^{-2}]}{[AT]}$$

$$\therefore R = \frac{[ML^2T^{-2} / AT]}{[A]} = [ML^2T^{-3}A^{-2}]$$

3. D

Sol. $T - 1 \times g = \left(\frac{5-4}{9} \right) g$

4. B

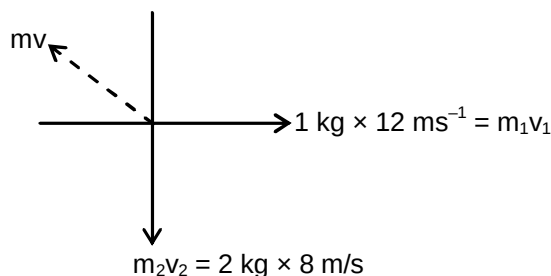
Sol. By work-energy theorem,
 $W = \Delta KE$

$$W_g + W_r = \frac{1}{2}mv^2$$

$$mgh + W_r = \frac{1}{2}mv^2 \quad \text{or} \quad W_r = \frac{1}{2}mv^2 - mgh$$

5. D

Sol. When an explosion breaks a rock, by the law of conservation of momentum, initial momentum which is zero, is equal to total momentum of three pieces.



Total momentum of the two pieces 1 kg and 2 kg

$$= \sqrt{12^2 + 16^2} = 20 \text{ kg ms}^{-1}$$

The third piece has the same momentum and in the direction opposite to the resultant of these two momenta.

∴ Momentum of third piece = 20 kg ms⁻¹ ; velocity = 4 ms⁻¹

∴ Mass of the 3rd piece = $\frac{mv}{v} = \frac{20}{4} = 5 \text{ kg}$.

6. D

$$\text{Sol. } T = \sqrt{\frac{4\pi^2 R^3}{GM}} \quad \text{and} \quad T' = \sqrt{\frac{4\pi^2 (2R)^3}{GM}}$$

$$\frac{T'}{T} = \sqrt{8} = 2\sqrt{2} \quad \text{or} \quad T' = 2\sqrt{2} T = 2.8T.$$

7. D

Sol. Poiseuille's formula gives the quantity of liquid flowing through a capillary,

$$Q = \frac{\pi p r^4}{8 \eta \ell}$$

$$\text{i.e., } p = \frac{8}{\pi} Q \cdot \frac{\eta \ell}{r^4}$$

$$\text{If } Q' = \frac{Q}{2}, \quad r' = 2r$$

$$\frac{p}{n} = \frac{8}{\pi} \frac{Q}{2} \cdot \frac{\eta \ell}{(2r)^4} = \frac{8}{\pi} \frac{Q \cdot \eta \ell}{r^4} \times \frac{1}{32} \quad (\because n = 32)$$

$$\text{i.e., pressure } p' = \frac{p}{32}.$$

8. B

Sol. $\sqrt{\frac{32}{17}}, V = \sqrt{\frac{rRT}{M}}, M_{\text{Mixture}} = \frac{2+32}{2} = 17$

9. A

Sol. $L_1 = 10 \log\left(\frac{I_1}{I_0}\right); L_2 = 10 \log\left(\frac{I_2}{I_0}\right)$

$$\therefore L_1 - L_2 = 10 \log\left(\frac{I_1}{I_0}\right) - 10 \log\left(\frac{I_2}{I_0}\right)$$

$$\text{or } \Delta L = 10 \log\left(\frac{I_1}{I_2}\right) \text{ or } 20 \text{ dB} = 10 \log\left(\frac{I_1}{I_2}\right)$$

$$\text{or } 10^2 = \frac{I_1}{I_2} \text{ or } I_2 = \frac{I_1}{100}.$$

10. B

Sol. $\sin \theta = \frac{1}{y n_x} = \frac{n_y}{n_x} = \frac{v_x}{v_y}$

$$\therefore v_y = \frac{v_x}{\sin \theta} = \frac{v}{\sin \theta}$$

11. C

Sol. $\sin i_c = \frac{1}{\mu} = \frac{3}{5}, \therefore \tan i'_c = \frac{3}{4}$

If x be the diameter of the disc, then

$$\tan i'_c = \frac{3}{4} = \frac{x}{4} \text{ or } x = 3 \text{ m}$$

12. B

Sol. $\frac{1}{f} = \frac{1}{f_1} = \frac{1}{f_2}$

$$\text{or } \frac{1}{30} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\therefore f \text{ is +ve, } \therefore |P_1| > |P_2|$$

Hence, $P_1 = 3, P_2 = 2$

$$\therefore \frac{f_1}{f_2} = \frac{P_2}{P_1} = \frac{2}{3}$$

$$\therefore f_1 = 10 \text{ cm and } f_2 = -15 \text{ cm}$$

13. C

Sol. For one prism, $\omega_1 = \frac{\delta_B - \delta_R}{\delta} = \frac{12 - 8}{10} \quad \left(\because \delta = \frac{\delta_B + \delta_R}{2} \right)$

or $\omega_1 = \frac{4}{10}$

For other prism, $\omega_2 = \frac{\delta_B - \delta_R}{\delta} = \frac{14 - 10}{12} \quad \left(\because \delta = \frac{14 + 10}{2} \right)$

or $\frac{\omega_1}{\omega_2} = \frac{4 \times 3}{10 \times 1} \quad \text{or} \quad \frac{\omega_1}{\omega_2} = \frac{12}{10} = \frac{6}{5}$

14. D

Sol. Current gain in common emitter mode

$$= \frac{\alpha}{1 - \alpha} = \frac{0.995}{1 - 0.995} = \frac{0.995}{0.005} = 199$$

15. D

Sol. Current I is given by:

$$20I + 10I = 2.5 - 1.0 \text{ or } I = \frac{1}{20} \text{ amp}$$

$$\therefore V_A - V_B = 2.5 - I \times 20 = 1.5 \text{ volt}$$

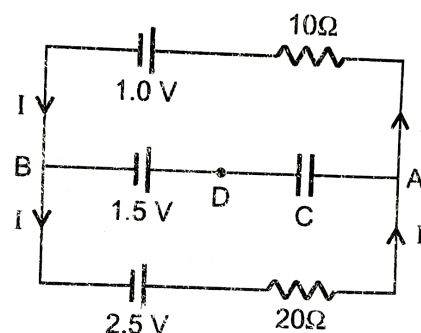
$$\text{But } (V_A - V_D) + (V_D - V_B) = V_A - V_B = 1.5$$

Since, there is no current through 1.5 volt cell,

so,

$$V_D - V_B = 1.5 \text{ volt}$$

$$\text{Hence, } V_A - V_D = 0 \text{ volt}$$



16. A

Sol. The resistance of 1 m length wire will be,

$$R = (40 \times 10^{-8}) / (8 \times 10^{-6}) = 5 \times 10^{-2} \Omega$$

$$\frac{\Delta V}{\Delta x} = (5 \times 10^{-2}) \times 0.2 = 10^{-2} \text{ V / m}$$

17. B

Sol. When the switch S is pressed, the circuit is complete but on account of electrical inertia of the inductor L, the current I_2 immediately after pressing the switch is zero, i.e., $I_2 = 0$.

Since, R_1 and R_2 are in series in the closed circuit, total resistance R is given by:

$$R = R_1 + R_2 = 2 + 3 = 5 \Omega$$

$$\therefore I_1 = I_3 = \frac{E}{R} = \frac{10}{5} = 2 \text{ amp.}$$

18. A

Sol. Here, $V_L = V_C$. They are in opposite phase. Hence, they will cancel each other. Now, resultant potential difference

= applied potential difference = 100 volt

$$Z = R \quad (\because X_L = X_C)$$

$$\therefore I_{\text{rms}} = \frac{V_{\text{rms}}}{R} = \frac{V_{\text{rms}}}{R} = \frac{100}{50} = 2 \text{ amp.}$$

19. B

Sol. λ (longest) for Balmer sides (36 / 5R) λ (longest) for Lyman sides = (4 / 3R)

$$\therefore \text{Ratio} = \frac{4}{3} \times \frac{5}{36} = \frac{5}{27}$$

20. D

$$\text{Sol. } \frac{1}{\lambda_{\alpha}} = R(Z-1)^2 \left(\frac{1}{1^2} - \frac{1}{2^2} \right)$$

$$\text{or } (Z-1)^2 = \frac{4}{3} \times \frac{1}{\lambda_{\alpha}} \times \frac{1}{R}$$

$$= \frac{4}{3} \times \frac{1}{1.785 \times 10^{-10}} \times \frac{10^{-2}}{109737}$$

$$\text{or } (Z-1)^2 = 680.6 \quad \text{or } (Z-1) = 26$$

$$\text{or } Z = 27$$

SECTION – B

21. 00002.50

$$\text{Sol. Poisson's ratio, } \sigma = 0.4 = \frac{\Delta d}{d} \ell \frac{\Delta \ell}{\ell}$$

$$\text{Area, } A = \pi r^2 = \frac{\pi d^2}{4} \quad \text{or } d^2 = \frac{4A}{\pi}$$

$$\text{Differentiation, } 2d \Delta d = \frac{4}{\pi} \cdot \Delta A$$

$$\text{As } A = \frac{\pi d^2}{4}, \text{ so, } \Delta A = \frac{2\pi d \Delta d}{4}$$

$$\therefore \frac{\Delta A}{A} = \frac{\frac{\pi d}{2} \Delta d}{\pi d^2 / 4} = 2 \frac{\Delta d}{d}$$

$$\text{Given; } \frac{\Delta A}{A} \times 100 = 2\%$$

$$\therefore 2 = 2 \frac{\Delta d}{d} \quad \text{or } \frac{\Delta d}{d} = 1\%$$

$$\text{Given; } \sigma = \frac{\Delta d / d}{\Delta \ell / \ell} = 0.4 \quad \text{or} \quad \frac{\Delta d}{d} = 0.4 \frac{\Delta \ell}{\ell}$$

$$\therefore \frac{\Delta \ell}{\ell} = \frac{1}{0.4} \frac{\Delta \ell}{\ell} = 2.5 \times 1\% = 2.5\%.$$

22. 00019.60

$$\text{Sol. Upthrust} = \frac{V}{2} \times 1000 \text{ g Newton}$$

$$\text{In equilibrium position } 2g = \frac{V}{2} \times 1000 \text{ g}$$

$$\therefore V = \frac{4}{1000} \text{m}^3$$

$\therefore$ Force required to wholly submerge the body into

$$\text{Water} = \frac{V}{2} \times d \times g = \frac{4 \times 1000 \times 9.8}{1000 \times 2} = 19.6 \text{ N.}$$

23. 00021.55

Sol. $\frac{E + \Delta E}{E} = \frac{[T + 95 / 100]T^4}{T^4}$

$$1 + \frac{\Delta E}{E} = (1 + 0.05)^4 = (1.05)^4$$

24. 00028.57

Sol. $C_p = \left(\frac{\gamma}{\gamma - 1} \right) R$

$$dQ = nC_p dT \quad \dots(i)$$

$$dU = nC_v dT$$

$$\therefore dW = dQ - dU = n(C_p - C_v)dT$$

$$= nR dT \quad \dots(ii)$$

Given: $dQ = 100 \text{ J}$

$$ndT = \frac{100}{C_p}$$

From equation (ii) $dW = R \left(\frac{100}{C_p} \right)$

$$= R \left(\frac{\gamma - 1}{\gamma R} \right) 100 = \left(\frac{1.4 - 1}{1.4} \right) \times 100 = 28.57 \text{ J.}$$

25. 00002.50

Sol. $E = \frac{100 - 95}{6 - 4} = \frac{5}{2} = 2.5$

26. 00000.02

Sol. For the first minimum, $a \sin \theta = \lambda$
where, a = width of slit

$$\therefore a = \frac{\lambda}{\sin \theta} = \frac{6980 \times 10^{-10}}{\sin 2^\circ} = 2 \times 10^{-5} \text{ m} = 0.02 \text{ mm.}$$

27. 00001.20

Sol. Internal resistance, $r = \frac{R(I_1 - I_2)}{I_2}$

$$= \frac{10(560 - 500)}{500} = \frac{10 \times 60}{500} = 1.2 \, \Omega$$

28. 00002.00

Sol. Comparing the given equation with the equation of plane electromagnetic wave,

$$E_z = E_0 \cos(\omega t + kx)$$

We have $\omega = 6 \times 10^8$ and $k = 4$

Velocity of light in medium,

$$v = \frac{\omega}{k} = \frac{6 \times 10^8}{4} = \frac{3}{2} \times 10^8 \text{ m/s}$$

$$\therefore \text{Refractive index, } \mu = \frac{c}{v} = \frac{3 \times 10^8}{\frac{3}{2} \times 10^8} = 2$$

29. 00003.33

Sol. The voltage drop across R_2 is

$$V_{R_2} = V_Z = 10 \text{ V}$$

The current through R_2 is

$$I_{R_2} = \frac{V_{R_2}}{R_2} = \frac{10 \text{ V}}{1500 \, \Omega} = 0.667 \times 10^{-2} \text{ A}$$

$$= 6.67 \times 10^{-3} \text{ A} = 6.67 \text{ mA}$$

The voltage drop across R_1 is

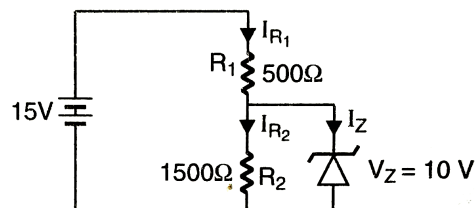
$$V_{R_1} = 15 \text{ V} - V_{R_2} = 15 \text{ V} - 10 \text{ V} = 5 \text{ V}$$

The current through R_1 is

$$I_{R_1} = \frac{V_{R_1}}{R_1} = \frac{5 \text{ V}}{500 \, \Omega} = 10^{-2} \text{ A} = 10 \times 10^{-3} \text{ A} = 10 \text{ mA}$$

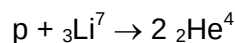
The current through the Zener diode is

$$I_Z = I_{R_1} - I_{R_2}$$



30. 00017.28

Sol. The reaction is,



Binding energy of

$${}_3\text{Li}^7 = 7 \times 5.60 = 39.2 \text{ MeV}$$

Binding energy of

$${}_2\text{He}^4 = 4 \times 7.06 = 28.24 \text{ MeV}$$

Energy of proton,

$$E_p = 2 \times \text{B.E.}({}_2\text{He}^4) - \text{B.E.}({}_3\text{Li}^7)$$

$$= 2 \times 28.24 - 39.2 = 17.28 \text{ MeV}$$

Chemistry

PART – B

SECTION – A

31. B

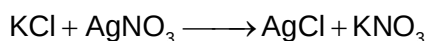
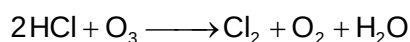
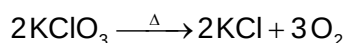
Sol. It is a zero order reaction.

32. C

Sol. Carbon should be present in the compound for giving nitrogen test.

33. C

Sol. $\text{NaCl} + \text{H}_2\text{SO}_4 \longrightarrow \text{NaHSO}_4 + \text{HCl}$



34. B

Sol. Roasting is carried out for sulphide ores.

35. C

Sol. It produces maximum number of ions as compared to other salts.

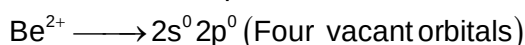
36. C

Sol. $\text{XY}_3(\text{g}) \longrightarrow \text{X}(\text{g}) + 3\text{Y}(\text{g}); \Delta H = E \text{ kJ mol}^{-1}$

$$\therefore \text{B.E of X - Y bond} = \frac{E}{3} \text{ as } \text{XY}_3 \text{ contains three X - Y bonds.}$$

37. A

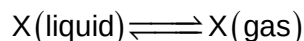
Sol. $\text{Be} \longrightarrow 1s^2 2s^2 2p^0$



$\therefore \text{Be}^{2+}$ forms complex with four water molecules.

38. D

Sol. At normal boiling point (500 K), the following equilibrium exists between the two physical states at 1 atm.



For the equilibrium $\Delta G = 0$

If the pressure increases the change shifts to backward direction for which $\Delta G = +\text{ve}$.

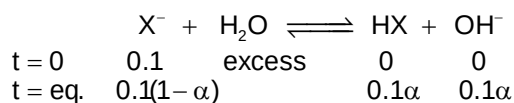
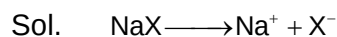
If pressure decreases the change shifts towards forward direction, for which $\Delta G = -\text{ve}$.

If temperature increase, the change will proceed towards forward direction, for which $\Delta G = -\text{ve}$.

39. C

Sol. CsCl has b.c.c. unit cell.

40. B



$$K_h = \frac{[\text{HX}][\text{OH}^-]}{[\text{X}^-]}$$

$$= \frac{[\text{HX}]}{[\text{H}^+][\text{X}^-]} \times \frac{[\text{H}^+][\text{OH}^-]}{1}$$

$$= \frac{K_w}{K_a} = \frac{0.1\alpha \times 0.1\alpha}{0.1(1-\alpha)} \approx 0.1\alpha^2 \quad [1-\alpha \approx 1 \text{ as } 1 \gg \alpha \text{ for weak acid}]$$

$$\therefore \frac{10^{-14}}{10^{-5}} = 0.1\alpha^2$$

$$\therefore \alpha = 1 \times 10^{-4}$$

$$\% \text{ hydrolysis} = 0.01\%$$

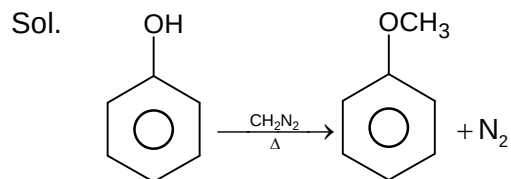
41. B

Sol. $\text{pOH} = \text{pK}_b + \log \frac{[\text{NH}_4^+]}{[\text{NH}_3]} = \text{pK}_b$

$$(\text{OH}^-) = 1.8 \times 10^{-5}$$

$$K_{sp} = [\text{Mn}^{2+}][\text{OH}^-]^2, [\text{Mn}^{2+}] = 1.38 \times 10^{-4}$$

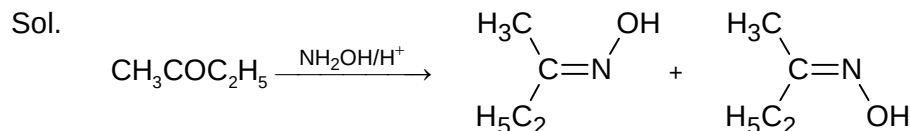
42. B



43. C

Sol. Anti Markownikoff reaction takes place in this case.

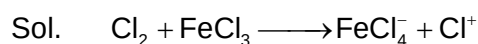
44. C



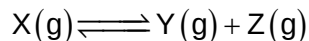
45. A

Sol. No Cl^- is formed in (A).

46. B



The electrophile Cl^+ substitutes a hydrogen atom from benzene.



At equ^m 1.25 1.25 1.25

After addition (1.25 + x - y) (1.25 + y) (1.25 + y)

$$1.25 + y = 1.3$$

$$\text{or, } y = 0.05$$

$$\therefore K_c = \frac{1.3 \times 1.3}{1.25 + x - 0.05} = 1.25$$

On solving, $x = 0.152$

$$0.152 = \frac{n}{V} = \frac{n}{100L}$$

$$\text{or, } n = 15.2$$

55. 00003.00

Sol. $\Delta H = nC_v \Delta T = 1 \times \frac{3}{2} R \times (1300 - 300) = \frac{3}{2} \times 2 \times 1000 = 3000 \text{ cal} = 3 \text{ Kcal.}$

56. 00003.00

Sol. $E_{\text{Cell}} = E_{\text{Cell}}^{\circ} - \frac{0.0591}{n} \log \frac{[Zn^{2+}]}{[H^+]^2}$

$$\text{or, } 0.6418 = [0 - (-0.76)] - \frac{0.0591}{2} \log \frac{10^{-2}}{[H^+]^2}$$

$$\text{on solving, } [H^+]^2 = 10^{-6}$$

$$\therefore [H^+] = 10^{-3}, \text{ pH} = 3$$

57. 00004.00

Sol. n-factor of NH_4NO_3 is 4.

58. 00004.00

Sol. It shows geometrical as well as optical isomerism. Optical isomerism will be exerted when one double bond of the diene is Cis and the other is trans.

59. 00003.00

Sol. Ortho, meta and para products are formed.

60. 00000.40

Sol. Moles of water = $\frac{72}{18} = 4$

$$\text{Total mole} = 4 + 6$$

$$\text{Mole fraction of water is } \frac{4}{10} = 0.4$$

Mathematics

PART – C

SECTION – A

61. C

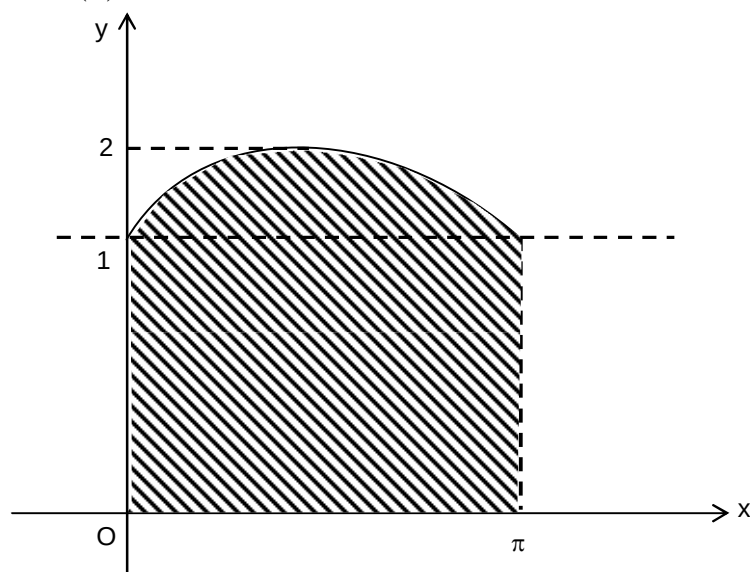
Sol. $y^2 - x^2 = 4 \Rightarrow 2y \frac{dy}{dx} - 2x = 0$
 $\Rightarrow y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 - 1 \Rightarrow y \frac{d^2}{dx^2} \frac{x^2}{y^2} - 1$
 $y^3 \frac{d^2y}{dx^2} - y^2 = x^2 - 4$

62. D

Sol. $C_1C_2 = r_1 + r_2$
 $C_1 = (0,0); C_2 = (3\sqrt{3}, 3)$ and $r_1 = 2, r_2 = 4$
 $\Rightarrow$ Circle touch each other externally.
 Equation of common tangent is $\sqrt{3}x + y - 4 = 0$ (i)
 Comparing it with $x \cos \theta + y \sin \theta = 2$, we get $\theta = \frac{\pi}{6}$

63. B

Sol. $f''(x) = 1 - f(x)$
 Multiplying both sides by $f'(x)$
 get $f(x) = \sin x + 1$ (i)



Required area = $\pi + 2$

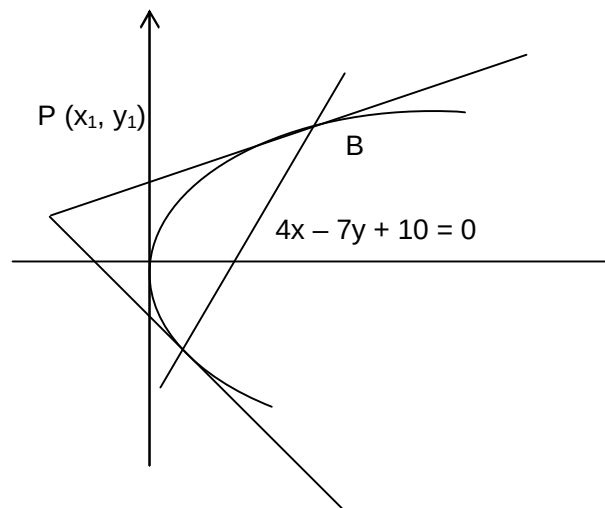
64. C

Sol. C.O.C. of $P(x_1, y_1)$

$$\text{w.r.t } y^2 = 4ax \text{ is } yy_1 = 2(x + x_1)$$

$$\text{compare with } 4x - 7y + 10 = 0$$

$$\text{to get } (x_1, y_1) = \left(\frac{5}{2}, \frac{7}{2}\right)$$



65. B

$$\text{Sol. } P(\omega) = 1 + a\omega^2 + b\omega + c = 0 \Rightarrow 1 + c - a + (b - a)\omega = 0$$

$$a - c = 1 \text{ and } b = a$$

$$\Rightarrow b = a \text{ and } c = a - 1$$

$$\Rightarrow \text{we can select in 9 ways.}$$

$$\Rightarrow P(E) = \frac{9}{1000}$$

66. B

$$\text{Sol. Given series is } \frac{3}{(1 \times 2)^2} + \frac{5}{(2 \times 3)^2} + \frac{7}{(3 \times 4)^2} + \dots$$

$$T_n = \frac{3 + (n-1) \times 2}{\{n(n+1)\}^2} = \frac{(n+1)^2 - n^2}{n^2(n+1)^2}$$

$$T_n = \frac{1}{n^2} - \frac{1}{(n+1)^2}$$

$$\therefore \text{Sum of series} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{2^2} - \frac{1}{3^2} + \dots + \frac{1}{n^2} - \frac{1}{(n+1)^2}$$

$$= 1 - \frac{1}{(n+1)^2} = \frac{n(n+2)}{(n+1)^2}$$

67. C

Sol. A is idempotent matrix i.e. $A^2 = A$

$$(I - 0.4A)^{-1} = I - \alpha A$$

$$\Rightarrow I = (I - \alpha A)(I - 0.4A)$$

$$= I^2 - 0.4AI - \alpha AI + 0.4\alpha A^2$$

$$= I + A(-0.4 - 0.6\alpha)$$

$$\Rightarrow \alpha = \frac{-4}{6}$$

68. A

Sol. $17^{1983} + 11^{1983} - 7^{1983}$

$$= (17^{1983} - 7^{1983}) + 11^{1983}$$

$$= (17 - 7)(17^{1982} + 17^{1981} \cdot 7^1 + 17^{1980} \cdot 7^2 + \dots + 7^{1982}) + 11^{1983}$$

$\therefore$ Last digit in $17^{1983} - 7^{1983}$ is 0
 and last digit in 11^{1983} is 1
 $\therefore$ Last digit of $17^{1983} + 11^{1983} - 7^{1983}$ is $0 + 1 = 1$

69. D

Sol.
$$\begin{vmatrix} \lambda - 1 & 3\lambda + 1 & 2\lambda \\ \lambda - 1 & 4\lambda - 2 & \lambda + 3 \\ 2 & 3\lambda + 1 & 3(\lambda - 1) \end{vmatrix} = 0$$

$$\Rightarrow \lambda = 0 \text{ or } 3$$

If $\lambda = 0$, the equations become $-x + y = 0$,

$$-x - 2y + 3z = 0 \text{ and } 2x + y - 3z = 0,$$

$$\therefore \frac{x}{6-3} = \frac{y}{6-3} = \frac{z}{-1+4}$$

70. C

Sol. 2^{nd} blue result occur on or before 10^{th} roll is equivalent to the occurrence of blue face at least twice in 10 rolls

$$n = 10, p = \frac{1}{3}, q = \frac{2}{3}$$

$$p(m \geq 2) = 1 - (p(0) + p(1)) = \frac{3^9 - 2^{11}}{3^9}$$

71. B

Sol. The equation of lines are $y - y_1 = \frac{m \pm \tan \alpha}{1 \mp m \tan \alpha} (x - x_1)$

$$\Rightarrow y - 4 = \frac{1 \pm \tan 45^\circ}{1 \mp \tan 45^\circ} (x - x_1)$$

$$\Rightarrow y - 4 = \frac{1 \pm 1}{1 \mp 1} (x - 3)$$

$$\Rightarrow y = 4 \text{ or } x = 3$$

72. B

Sol. $(x-2)^2 + (y+3)^2 = \left(\frac{1}{2} \frac{3x-4y+7}{5}\right)^2$ is an ellipse, whose focus is (2, -3), directrix

$3x - 4y + 7 = 0$ and eccentricity is $\frac{1}{2}$.

Length of $\perp$ from focus to directrix is $\frac{3 \times 2 - 4 \times (-3) + 7}{5} = 5$

$$\frac{a}{e} - ae = 5$$

$$\Rightarrow 2a - \frac{a}{2} = 5 \Rightarrow a = \frac{10}{3}$$

So length of major axis is $\frac{20}{3}$

73. C

Sol. $ax^3 + bx + c = (x^2 + px + 1)(ax + c)$ comparing co-eff of x^2 and x , we get $p = \frac{-c}{a}$

$$\Rightarrow a + c\left(\frac{-c}{a}\right) = b \Rightarrow a^2 - c^2 = ab$$

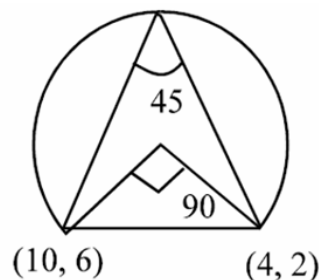
74. D

$$\text{Sol. } \sqrt{2r} = \sqrt{(10-4)^2 + (6-2)^2}$$

$$= \sqrt{36 + 16} = 2\sqrt{13}$$

$$r = \sqrt{26}$$

$$\text{Perimeter} = \frac{3}{4} \times 2\pi r = \frac{3}{2} \pi \sqrt{26}$$



75. B

Sol. Clearly sum of digits = 22, for the number to be divisible by 9 the sum of digits must be divisible by 9 $\Rightarrow$ sum of digit is 18 only.

There are two ways (1) omit 0 and 4

Number of arrangements = $5! = 120$.

(2) By omitting 1, 3

Then the number of ways = $4 \times 4 \times 3 \times 2 \times 1 = 96$.

$\therefore$ Total number of numbers = $120 + 96 = 216 = 6^3$.

76. A

Sol. Given, $\sin \beta = \sqrt{\sin \alpha \cos \alpha} \Rightarrow \sin^2 \beta = \sin \alpha \cos \alpha$

$$\text{Now, } \cos 2\beta = 1 - 2\sin^2 \beta = 1 - 2\sin \alpha \cos \alpha$$

$$= (\sin \alpha - \cos \alpha)^2$$

$$= 2 \sin^2 \left(\frac{\pi}{4} - \alpha \right) \text{ or } 2 \cos^2 \left(\frac{\pi}{4} + \alpha \right)$$

77. B

Sol. The given equation is equivalent to $2 \sin^2 \left(\left(\frac{\pi}{2} \right) \cos^2 x \right) = 2 \sin^2 \left(\left(\frac{\pi}{2} \right) \sin 2x \right)$

$$\Rightarrow \cos^2 x = \sin 2x$$

$$\Rightarrow \cos x (\cos x - 2 \sin x) = 0$$

$$\Rightarrow 1 - 2 \tan x = 0 \text{ as } \cos x \neq 0$$

$$x \neq (2n+1) \frac{\pi}{2}$$

$$\Rightarrow \tan x = \frac{1}{2}$$

$$\Rightarrow \cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x} = \frac{3}{5}$$

78. A

Sol. $\sin^{-1} \left(\sqrt{\frac{2-\sqrt{3}}{4}} \right) = \sin^{-1} \left(\sqrt{\frac{4-2\sqrt{3}}{8}} \right)$

$$= \sin^{-1} \left(\sqrt{\frac{(\sqrt{3}-1)^2}{8}} \right) = \sin^{-1} \left(\frac{\sqrt{3}-1}{2\sqrt{2}} \right) = \frac{\pi}{12}$$

$$\cos^{-1} \left(\frac{\sqrt{12}}{4} \right) = \cos^{-1} \left(\frac{2\sqrt{3}}{4} \right) = \frac{\pi}{6}$$

$$\sec^{-1}(\sqrt{2}) = \frac{\pi}{4}$$

$$\cos^{-1} \left[\cot \left\{ \frac{\pi}{12} - \frac{\pi}{6} + \frac{\pi}{4} \right\} \right] = \cos^{-1} \left[\cot \left\{ \frac{\pi}{2} \right\} \right]$$

$$= \cos^{-1}(0) = \frac{\pi}{2}$$

79. C

Sol. $f(g(x)) = 1 + 3 \left(\frac{x - 2x^2 + 5x^3}{3} \right)$

$$\Rightarrow g(x) = \frac{5x^3 - 2x^2 + x}{3}$$

$$\therefore 3g(1) = 4$$

80. A

Sol.
$$\lim_{n \rightarrow \infty} \sum_{r=0}^n \frac{{}^nC_r}{n^r} \left(\int_0^1 x^{r+3} dx \right)$$

$$= \int_0^1 \left(\lim_{n \rightarrow \infty} \sum_{r=0}^n {}^nC_r \cdot \left(\frac{x}{n} \right)^r \cdot x^3 \right) \cdot dx$$

$$= \int_0^1 \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n \cdot x^3 dx$$

$$= \int_0^1 e^x \cdot x^3 dx :$$

$$= 6 - 2e$$

SECTION - B

81. 00005.00

Sol.

$$\lim_{n \rightarrow \infty} \frac{\int_0^{\frac{1}{n}} x^{x+k-1} dx}{\left(\frac{1}{n} \right)^k} = \lim_{t \rightarrow 0} \frac{\int_0^t x^{x+k-1} dx}{t^k}$$

Taking t as continuous variable (as $t \rightarrow 0$)

$$\lim_{n \rightarrow \infty} \frac{t^{t+k-1}}{k t^{k-1}} = \lim_{t \rightarrow 0} \frac{1}{k} t^t = \frac{1}{k}.$$

82. 00005.00

Sol. Equation of tangent to parabola at P is

$$2(y+9) = 6x \Rightarrow y+9 = 3x$$

Equation of circle is

$$(x-6)^2 + (y-9)^2 + \lambda(y-3x+9) = 0$$

Put (0, 1)

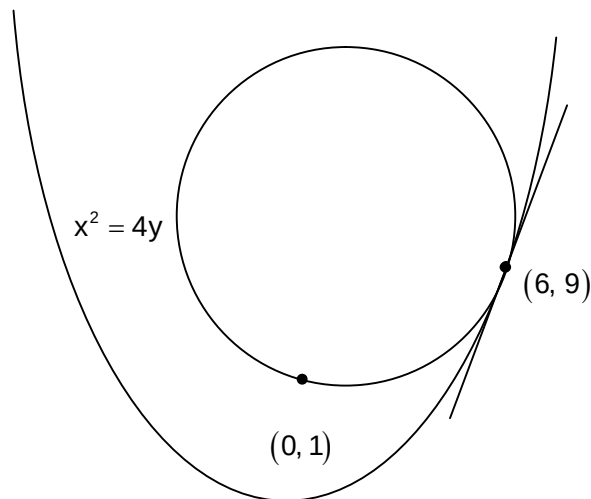
$$\Rightarrow 36 + 64 + \lambda(10) = 0$$

$$\Rightarrow \lambda = -10$$

 $\therefore$ Equation of circle is

$$x^2 + y^2 + 18x - 28y + 27 = 0$$

$$\therefore \text{radius} = \sqrt{9^2 + (14)^2 - 27} = 5\sqrt{10}$$



83. 00005.00

Sol. $y' \ln x + \frac{y}{x} = \ln y + \frac{x}{y} y'$ for the point (e^2, e^2)

$$2y' + 1 = 2 + y'$$

$$y' = 1$$

$$\text{Equation of normal is } y - e^2 = -1(x - e^2)$$

$$y + x = 2e^2$$

$$a = 2e^2 \text{ and } b = 2e^2$$

$$\left| \frac{4a+b}{b} \right| = 5$$

84. 00003.00

Sol. Equation of normal at $(2 \sec \theta, \tan \theta)$ is $\frac{2x}{\sec \theta} + \frac{y}{\tan \theta} = 5$

$$\text{Slope} = -1 \Rightarrow -\frac{2 \tan \theta}{\sec \theta} = -1$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

$$\therefore \text{normal becomes } y = -x + \frac{5}{\sqrt{3}}$$

$$\text{It is tangent to ellipse} \Rightarrow a^2 + b^2 = \frac{25}{3}$$

$$\Rightarrow \frac{9}{25}(a^2 + b^2) = 3$$

85. 00002.00

Sol. $f(x)$ is differentiable in $\left(0, \frac{1}{\sqrt{2}}\right)$

$\therefore f(x)$ is continuous in $\left[0, \frac{1}{\sqrt{2}}\right]$

$$\Rightarrow f\left(\frac{1}{2}\right) \Rightarrow \lim_{x \rightarrow \frac{1}{2}} f(x) \Rightarrow \lim_{x \rightarrow \frac{1}{2}} \tan^{-1}\left(\frac{ax+b}{c}\right) = 0 \text{ and } \lim_{x \rightarrow \frac{1}{2}} \log(2+bx^2) = 0$$

$$\Rightarrow \frac{a}{2} + b - 0 \text{ and } 2 + \frac{b}{4} - 1$$

$$\Rightarrow a + 2b = 0 \text{ and } b = -4 \Rightarrow a = 8$$

Now, $\therefore f(x)$ is differentiable at $x = \frac{1}{2}$

$$\Rightarrow \frac{1}{1 + \left(\frac{ax+b}{c}\right)^2} \frac{a}{c} \Big|_{x=\frac{1}{2}} - \frac{1}{2+bx^2} 2bx \Big|_{x=\frac{1}{2}}$$

$$\Rightarrow \frac{8}{c} - \frac{-4}{2 - 4\left(\frac{1}{4}\right)} \Rightarrow c = -2$$

86. 00009.00

Sol.
$$\frac{\frac{7}{2}[2a+6d]}{\frac{11}{2}[2a+10d]} = \frac{6}{11} \Rightarrow \frac{7(2a+6d)}{(2a+10d)} = 6$$

$$\Rightarrow 2a = 18d$$

$$a = 6d$$

also $130 < a + 6d < 140$

$$\frac{26}{3} < d < \frac{28}{3} \Rightarrow d = 9$$

87. 00001.71

Sol. The given series is in the form $1 + a + a^2 + a^3 \dots$ to ∞ where $a = \left(\frac{\sqrt{2}-1}{2\sqrt{2}} \right)$

$$S_n = \frac{1}{1-a}$$

$$= \frac{1}{1 - \left(\frac{\sqrt{2}-1}{2\sqrt{2}} \right)}$$

$$= \frac{2\sqrt{2}}{2\sqrt{2} - \sqrt{2} + 1}$$

$$= \frac{2\sqrt{2}}{\sqrt{2} + 1}$$

$$= 2\sqrt{2}(\sqrt{2} - 1)$$

88. 00009.00

Sol.
$$\int \frac{2x^7 + 3x^2}{x^{10} - 2x^5 + 1} dx$$

$$= \int \frac{x^6 \left(2x + \frac{3}{x^4} \right)}{x^6 \left(x^4 - \frac{2}{x} + \frac{1}{x^6} \right)} dx$$

$$= \int \frac{2x + \frac{3}{x^4}}{\left(x^2 - \frac{1}{x^3} \right)^2} dx$$

Let $x^2 - \frac{1}{x^3} = t \Rightarrow \left(2x + \frac{3}{x^4} \right) dx = dt$

$$= \int \frac{dt}{t^2} = -\frac{1}{t} + c = -\frac{x^3}{x^5 - 1} + c$$

$$= \frac{-x^3(x^5 - 1)}{(x^5 - 1)^2} + c = \frac{x^3 - x^8}{(x^5 - 1)^2} + c$$

89. 00003.00

Sol. $f(x) = x^3 + bx^2 + cx$

$$f(1) = 1 + b + c$$

$$f(2) = 8 + 4b + 2c$$

By Rolle's Theorem

$$f(1) = f(2)$$

$$\Rightarrow 3b + c + 7 = 0 \quad \dots\dots(i)$$

$$f'(x) = 3x^2 + 2bx + c$$

$$f'\left(\frac{4}{3}\right) = 0$$

By Rolle's theorem

$$\Rightarrow 8b + 3c + 16 = 0 \quad \dots\dots(ii)$$

By (i) and (ii)

$$\frac{b}{-5} = \frac{c}{8} = \frac{1}{1}$$

$$b = -5$$

90. 00001.00

Sol. $\therefore \frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$

$$\begin{aligned} \frac{d^2x}{dy^2} &= \frac{d}{dy} \left(\frac{1}{\frac{dy}{dx}} \right) = \frac{d}{dx} \left(\frac{1}{\frac{dy}{dx}} \right) \cdot \frac{dx}{dy} \\ &= \frac{-1}{\left(\frac{dy}{dx} \right)^2} \cdot \frac{d^2y}{dx^2} \cdot \frac{dx}{dy} = \frac{-1}{\left(\frac{dy}{dx} \right)^3} \cdot \frac{d^2y}{dx^3} \end{aligned}$$

$$\text{From the given is } \frac{d^2y}{dx^2} \left(1 - \frac{1}{\left(\frac{dy}{dx} \right)^3} \right) = 0$$

$$\Rightarrow \frac{d^2y}{dx^2} = 0 \text{ or } \frac{dy}{dx} = 1$$

$$\Rightarrow y = ax + b : a \neq 0$$